

**POLA VIGNESH**Course : **B.Tech**, Electronics & Communication Engineering, 2026

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CGPA : 8.88



ACADEMIC DETAILS					
COURSE	SPECIALIZATION	INSTITUTE/COLLEGE	BOARD/UNIVERSITY	SCORE	YEAR
DIP	Electronics & Communication Engineering	Government polytechnic college, Masab Tank	State Board of Technical Education and Training	9.8 CGPA	2023
CLASS X		VAMSHI HIGH SCHOOL	Telangana Board of Secondary Education (SSC)	10 CGPA	2020

Subjects / Electives	Electronic Devices and Circuits, Microprocessor and Microcontroller, Analog and Digital Communications, Digital Signal Processor, Digital Logic Design, Antenna Wave Propagation, Satellite Communications, Network Analysis, Microwave and Optical Fibre Communication, Integrated Circuits, VLSI, Signals and Systems
Technical Proficiency	MS Word, C Programming, Programming, Assembly Language, MATLAB, ANSYS, Problem Solving, Cadence Virtuoso, Excel, Keil uVision, Presentation Skills, AutoCAD, Aptitude, Python3, Java, SQL, Analog Circuits, Communication Skills, Embedded C, Boolean Logic, ARM7, Linux, Arduino

WORK EXPERIENCE	
trainee, SPR HCS IN MY DIPLOMA LAST SEMESTER, I HAVE DONE THE INDUSTRIAL TRAINING IN THIS COMPANY UNDER OPTICAL FIBRE AND COMMUNICATIONS	Jan 2022 - May 2023

INTERNSHIPS	
Real time programming using using QNX, Pi Square Technologies - Gained hands-on understanding of the QNX architecture and real-time operating system concepts. - Learned about microkernel-based OS design, process management, and inter-process communication (IPC). - Explored QNX development tools and practiced basic system configuration and debugging. - Developed foundational knowledge of embedded systems used in automotive and industrial applications	Aug 2025 - Oct 2025
Automotive Embedded Engineer, Training - During my internship, I had the opportunity to work extensively with CANoe, a comprehensive software tool developed by Vector Informatik. - CANoe is widely used in the automotive industry for the development, simulation, testing, and analysis of in-vehicle communication networks such as CAN, LIN, FlexRay, and Ethernet. - Through this experience, I gained practical knowledge of how CANoe facilitates the design and validation of embedded systems in automotive applications. - I was involved in configuring network nodes, analyzing communication protocols, and simulating real-time vehicle behavior to test ECU functionalities. - This exposure significantly enhanced my understanding of automotive communication systems and the role of tools like CANoe in ensuring system reliability and performance."	May 2024 - Jun 2024
Embedded Systems, Launched - Completed hands-on internship in Embedded Systems, working on microcontroller-based system design and development. - Designed and tested embedded applications using C programming and microcontroller platforms . - Interfaced peripherals such as sensors, LCDs, motors, and communication modules (UART, I2C, SPI). - Developed and debugged firmware for real-time embedded applications. - Gained practical experience with hardware components, circuit design, and PCB basics. - Performed troubleshooting using tools like multimeters, oscilloscopes, and simulation software. - Worked on mini-projects involving automation/IoT-based systems (customize based on your project). - Understood concepts of RTOS, interrupt handling, and embedded communication protocols. - Collaborated with a technical team to test, validate, and optimize system performance	Sep 2025 - Oct 2025
Embedded Systems, Practical Eduskills - During this internship, I gained hands-on experience in the design, development, and testing of embedded systems. - My responsibilities included programming microcontrollers using ALP and C language, interfacing sensors and actuators, and working with embedded development tools like Keil and Tina. - I contributed to building real-time embedded applications, debugging hardware-software integration issues, and optimizing system	Jul 2024 - Sep 2024

performance.

PROJECTS

Illustration Of Thread Synchronisation in Intelligent Traffic Light Controller using Condvars - Embedded System

Nov 2025 - Dec 2025

- Implemented thread synchronization using condition variables to coordinate communication between multiple threads.
- Demonstrated concepts of mutual exclusion, thread waiting, and signaling mechanisms.
- Used **POSIX Threads (pthreads)** condition variables to manage shared resources and avoid race conditions.
- Improved understanding of concurrency control and efficient thread management in multithreaded applications.

Monitoring and Controlling of Green House Environment - Embedded System

Aug 2022 - Oct 2022

- The greenhouse environment monitoring project is designed to track and regulate key environmental parameters such as temperature, humidity, soil moisture, and light intensity to ensure optimal conditions for plant growth.
- Using sensors connected to a microcontroller such as Arduino , real-time data is collected and analyzed.
- This data can be displayed on an LCD or transmitted to a remote monitoring system.
- The system may also include automation features, such as triggering fans, irrigation, or lighting systems based on predefined thresholds.
- This project helps in improving crop yield, reducing manual intervention, and promoting sustainable agricultural practices."

Smart Power Management and Safety Load Management - Automotive Embedded system

Sep 2024 - Nov 2024

- In automotive systems, efficient power management and safety are paramount to ensuring optimal vehicle performance, preventing electrical hazards, and enhancing the overall user experience.
- Traditional power distribution systems often lack the intelligence required to dynamically manage power loads or effectively respond to safety-critical events such as overloads, short circuits, or system faults.
- This limitation can lead to reduced efficiency, potential system failures, and compromised safety."

Analog clock using Matlab - Matlab

Mar 2024 - Apr 2024

- An analog clock using MATLAB is a graphical simulation that visually represents the current time with hour, minute, and second hands, similar to a traditional clock.
- This application utilizes MATLAB's graphical functions to draw the clock face and dynamically update the hand positions based on the system time.
- By leveraging MATLAB's plotting capabilities and real-time data processing, the analog clock provides an interactive and educational tool for understanding time-based graphical programming and GUI development."

Monitoring of heart beat using Arduino - Embedded System

Mar 2024 - Apr 2024

- "A heart rate monitoring system using Arduino is a compact, low-cost solution designed to measure a person's heart rate in real-time.
- The system typically uses a pulse sensor, which detects the heartbeat by sensing changes in blood volume through the fingertip or earlobe.
- The sensor sends analog signals to the Arduino microcontroller, which processes the data to calculate beats per minute (BPM).
- The results can be displayed on an LCD screen or transmitted to a computer or mobile device for further analysis.
- Such systems are useful in health monitoring, fitness tracking, and biomedical applications due to their simplicity, portability, and affordability."

COMPETITIONS

HackWave

Nov, 2024

- A **CNC plotter** is a precision-driven, computer-controlled machine designed to reproduce complex vector-based drawings or patterns on various flat surfaces.
- Utilizing **Computer Numerical Control (CNC)** technology, it interprets digital design files—typically in G-code or similar formats—to guide a tool (such as a pen, engraver, or cutter) along defined X, Y, and sometimes Z axes.
- CNC plotters are widely employed in fields such as **engineering design, electronics prototyping and manufacturing**, offering high levels of accuracy, repeatability, and automation. Depending on the toolhead , they can perform a range of tasks including plotting, engraving, cutting, and marking, making them a versatile asset in both industrial and educational settings.

CONFERENCES AND WORKSHOPS

VLSI workshop

Organized by: **BITS PILANI HYDERABAD** | Date: **Apr 2025**

- The VLSI (Very-Large-Scale Integration) workshop is designed to provide participants with foundational and practical knowledge in the

field of integrated circuit design.

- The workshop covers key aspects of VLSI technology, including digital and analog circuit design, CMOS technology, HDL (Hardware Description Language) programming, FPGA implementation

LANGUAGES KNOWN

English, Hindi, Telugu